25900358

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEECT200 - INTRODUCTION TO EMBEDDED SYSTEM

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. AVR microcontrollers are based on which architecture? [K] / [CO1]

   a). CISC architecture Ans. **b). Harvard architecture**

   c). Super sclar architecture d). Von Neumann architecture

2. What is the function of the AREF pin? [K] / [CO2]

   a). Analog input b). Digital output

   Ans.c**). Reference voltage for ADC** d). Analog reset

3. The _delay_ms() function is used to create a delay in [K] / [CO3]

   Ans.a). Milliseconds b). Microseconds

   c). Minutes d). Clock cycles

4. If DDRB = 0x01;, what does it mean? [A] / [CO4]

   a). PB0 is input, all others output b). All pins are output

Ans.c). PB0 is output, all others input d). All pins are input

5. Which is a specialized controller optimized for tasks like filtering and compression? [U] / [CO1]

   Ans.a). Digital Signal Processor b). GPU

   c). Microcontroller d). Microprocessor

6. Which characteristics typically do not represent a RISC processors? [K] / [CO1]

   a). Simple instructions **Ans.b). Less registers**

   c). Fixed instruction format d). Single clock cycle executiion

7. How many interrupt sources does ATmega32 have? [K] / [CO2]

   a). 14 **Ans.b). 21**

   c). 32 d). 40

8. What is the purpose of #include in an AVR Embedded C program? [K] / [CO3]

   a). To access math functions

   b). . To define delay functions

   c). **To access I/O register definitions :Ans**

   d). To include standard I/O streams

9. Why is a loop of 8 iterations needed to count ones in a byte? [A] / [CO4]

   a). Because the AVR can only handle 8 instructions per loop

   b). **Because each iteration checks one bit of the 8-bit register: Ans**

   c). Because ROR automatically clears bits after 8 rotations

   d). Because the Carry flag resets after 8 shifts

10. In the AVR interrupt priority scheme, which interrupt is served first if two occur simultaneously? [U] / [CO3]

    a). The one with the higher vector address

b). **The one with the lower vector address: Ans**

c). The one enabled last

d). The one with the higher I bit

11. Flash memory in AVR microcontrollers is used to store [K] / [CO1]

a). **Program Code**: Ans b). Temporary values

c). External fies d). Sensor data

12. Which flag does BRNE rely on? [K] / [CO2]

a). Carry Ans: b**). Zero**

c). Negative d). Overflow

13. Which data type is most efficient for storing a value between 0 and 100 in AVR? [U] / [CO3]

Ans a). unsigned char b). float

c). char d). int

14. To extend 16 bit subtraction to program for 32-bit subtraction, which change would be required? [U] / [CO4]

a). Use SUB for all bytes

b). **Use SUB for least significant byte and SBC for all higher bytes :Ans**

c). Replace BRCC with RJMP

d). Remove the LDI instruction entirely

15. Which addressing mode uses the Program Counter (PC) relative displacement for branch instructions? [K] / [CO2]

a). Direct addressing b). Immediate addressing

c**). Relative addressing: Ans** d). Indirect addressing

16. What is the effect of the ROR instruction? [K] / [CO2]

Ans: a**). Rotate right through carry** b). Rotate right without carry

c). Clear register d). Copy right bit

17. Which flag is set when an arithmetic carry occurs from bit 3 to bit 4? [K] / [CO2]

a). C b). Z

Ans: c). H d). N

18. In an AVR project, an engineer wants to configure PORTC as output, using Embedded C. What is the correct code? [A] / [CO3]

Ans: a). DDRC= 0xFF; b). PORTC = 0xFF;

c). PORTC = 0x00; d). PINC = 0xFF;

19. Which instruction is used to load an immediate value into a register in AVR assembly? [K] / [CO4]

a). LDS **Ans:b). LDI** c). ADD d). STS

20. Single-purpose embedded processors also known as [U] / [CO1]

a). Microprocessor b). CPU

c). DSP Ans: **d). ASICs**

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

21. Discuss why microcontrollers are preferred for embedded systems [U] / [CO1]

Integration of components, low cost, low power consumption, real time operations, compact size, reliability and stability, versality, easy to program and interface etc.(Short description)

22. Which instruction is used to store data from a register to a SRAM location pointed by an address register? Explain. [K] / [CO2]

The instruction used to store data from a register to an SRAM location pointed by an address register in AVR microcontrollers is the ST (Store Indirect) instruction.

ST:Stores data from a general-purpose register into the data memory location pointed to by the X, Y, or Z register pair. For example,

ST Y+, R21 stores the value of register R21 into the memory location pointed to by the Y register pair and then increments the Y register.

23. Write an AVR C program to get a byte of data from PORTB and then send it to PORT C [U] / [CO3]

```
#include <avr/io.h>

int main(void)

{

   DDRB = 0x00;   // Configure PORTB as input (0 = input)

   DDRC = 0xFF;   // Configure PORTC as output (1 = output)

   while (1)

   {

      PORTC = PINB;   // Read data from PORTB and output it to
PORTC

   }

   return 0;

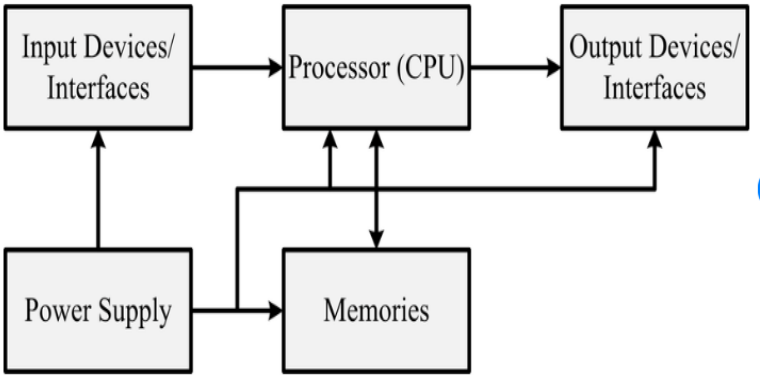
}
```

24. Using an example, explain how a loop is used to implement multiplication by repeated addition. [A] / [CO4]

```
LDI R16,0x00          ; Clear R16 (will hold the LOW byte of the product)
      LDI R17,0x00          ; Clear R17 (will hold the HIGH byte of the product)
      LDS R0,0x60           ; Load first operand (multiplicand) from SRAM address 0x60
      LDS R1,0x61           ; Load second operand (multiplier) from SRAM address 0x61
L2:   ADD R16,R0            ; Add multiplicand (R0) to result low byte (R16)
      BRCC L1               ; If no carry from ADD, skip increment of high byte
      INC R17               ; If carry occurred, increment high byte (R17)
L1:   DEC R1                ; Decrement multiplier (R1)
      BRNE L2               ; Repeat loop until multiplier reaches zero
```

25. What are the various hardware functional blocks required to build a typical embedded system? [K] / [CO1]

    Draw the block diagram and give short explanation



General block diagram of embedded systems.

26. Name the three categories of the AVR microcontroller family. Mention one key feature of each type. [K] / [CO2]

The AVR family of microcontrollers is typically classified into several groups based on their features, target applications, and footprint.

These groups include: Classic AVR,TinyAVR, MegaAVR, XmegaAVR, and sometimes Special Purpose AVRs.

Give one key feature of each type.

27. What does the AND instruction perform in AVR? Explain with the help of an appropriate example. [K] / [CO2]

Explain AND instruction with a sample program

28. Write the basic purposes of TCNTn, TCCRn, and TOVn registers? [K] / [CO3]

Basic registers of timers

1. TCNTn

In AVR, for each of the timers, there is a TCNTn (timer/counter) register. That means in ATmega32 we have TCNTO, TCNT1, and TCNT2. The TCNTn register is a counter. Upon reset, the TCNTn contains zero. It counts up with each pulse. The contents of the timers/counters can be accessed using the TCNTn. You can load a value into the TCNTn register or read its value.

2. TOVn

Each timer has a TOVn (Timer Overflow) flag, as well. When a timer overflows, its TOVn flag will be set.

3. TCCRn

Each timer also has the TCCRn (timer/counter control register) register for setting modes of operation. For example, you can specify Timer0 to work as a timer or a counter by loading proper values into the TCCRO.

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

29. Explain the basic architecture of a microprocessor system with the help of a block diagram. [K] / [CO1]

Block diagram -5 marks ,explanation- 5 marks

30. Describe structure of an AVR assembly language instructions. Briefly explain instruction categorization by function, with appropriate examples. [K] / [CO2]

An AVR assembly language instruction typically consists of an opcode (or mnemonic), which specifies the operation, and operands, which provide the data or addresses the operation acts upon. Instructions can also include labels for referencing specific memory locations and comments for clarity.

An Assembly language instruction consists of four fields:

```
[ label:]    mnemonic  [ operands]  [ ;comment]
```

1.Data transfer Instructions

2.Branching instruction

3.Bit manipulation instructions

4.Arithmetic and Logic instructions

Explain with appropriate examples.

31. Explain the features of embedded C. Describe the basic structure of a AVR C program, with a suitable example [K] / [CO3]

Ease of Programming:

C is easier to learn and use than assembly language, while still providing the necessary control over hardware.

Easy to modify and update

Low-Level Access:

C allows direct manipulation of hardware registers and peripherals, which is essential for embedded systems programming.

Portability:

Code written in C can be easily ported to different AVR devices and even other microcontroller families, with minimal modifications.

Efficiency:

C compilers can generate efficient machine code for AVRs, enabling fast execution and optimal resource utilization.

Community Support:

AVR microcontrollers have a large and active community, providing ample resources, libraries, and support for C programming.

Wide Range of Applications:

AVR microcontrollers are used in diverse applications, including IoT devices, home automation, robotics, and industrial automation, all of which often rely on Embedded C for their functionality.

&Explain the basic structure with suitable embedded C program

32. Explain with a neat diagram and Embedded C program, how an AVR microcontroller can control multiple LEDs in a sequence (LED chaser). [A] / [CO4]

[10x2 = 20]

Write program in embedded c for controlling a chaser -6 marks

Diagram- 4 marks